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\documentclass[11pt]{article}\usepackage[utf8]
{inputenc}\usepackage{amsmath,amssymb,amsthm}\usepackage{geometry}\usepackage{hyperref}
\geometry{a4paper, margin=1in}

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Technical Note: Causal Inference Logic in ADPL

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1. Introduction

The **Algorithmic-Discrimination-Proof-Ladder (ADPL)** framework addresses the "Black Box" challenge in legal proceedings[cite: 36]. Traditional statistical parity fails to account for neutral confounding factors[cite: 37]. This document derives the causal bias metric used in Phase 2 of the system[cite: 38].

2. The Causal Framework

We adopt the *Potential Outcomes* framework (Rubin-Causal Model)[cite: 40]. Let X be the protected attribute (e.g., gender), Y be the binary decision outcome, and W be a set of neutral covariates (e.g., education, experience)[cite: 41].

2.1. Defining the Counterfactual

The "But-for" test in legal theory corresponds to the *Individual Causal Effect*[cite: 43]:

$$\tau_i = Y_i(X = 1) - Y_i(X = 0) \quad (1)$$

Where $Y_i(1)$ is the outcome if the individual were female, and $Y_i(0)$ if they were male, holding all other traits constant[cite: 46].

2.2. Average Causal Effect (ACE)

To quantify systemic bias, we calculate the Average Causal Effect using Pearl's *do*-calculus:

$$ACE = E[Y|\text{do}(X = 1)] - E[Y|\text{do}(X = 0)] \quad (2)$$

3. Identification via Back-door Adjustment

Since W (Education) affects both X (Sampling) and Y (Decision), it acts as a confounder[cite: 50]. To isolate the causal link $X \rightarrow Y$, we adjust for W using the Back-door Criterion[cite: 51]:

$$P(Y|\text{do}(x)) = \sum_w P(Y|X = x, W = w)P(W = w) \quad (3)$$

4. Legal Threshold for Burden Shifting

In the ADPL system, the burden of proof shifts when the calculated Net Causal Bias exceeds the legal tolerance level α [cite: 54].

Decision Rule: The technical rebuttal is deemed insufficient and the burden of proof shifts if:

$$|ACE| > \alpha \quad (4)$$

Where $\alpha = 0.05$ is the established threshold for systemic discrimination[cite: 58]. If this condition is met, the “But-for” causation is legally established[cite: 59].

References

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